

Cooperation or Defection? The Decay of Alignment Instructions in Complex LLM Systems

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As autonomous LLM agents are increasingly deployed in social simulations and economic contexts, determining whether these systems possess the intrinsic capacity to cooperate in complex systems setting is of extreme relevance.

In this study, we constructed a computational laboratory designed to experiment with LLMs in repeated game theory scenarios over multiple periods, across various games and strategic configurations. Specifically, we focused on the classic Iterated Prisoner’s Dilemma, and the experimental design evaluated agents under two distinct conditions: a ”pure” behavior setting (agents operating without specific initial behavioral instructions) and a strategic setting (agents operating under pre-defined instructional prompts), both tested across a diverse array of models.

The results suggest that, while initial instructions exert a demonstrable effect on the system’s early behavior, this influence is observed to be moderated by subsequent interactions, suggesting that interactions may eventually override initial alignment prompts. Also, we observed heterogeneity in outcomes across different models, diverging toward different equilibria. These findings highlight the critical importance of training data and architectural bias when employing LLMs in complex systems.

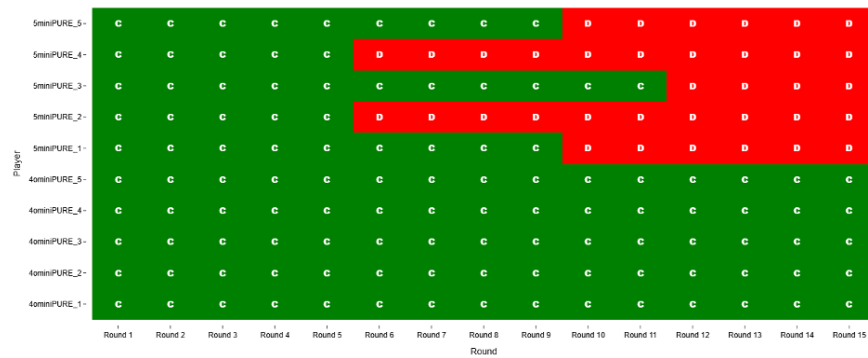


Figure 1: Example of progressive shifting of the system behaviour from a cooperation equilibrium from a defecting one